

Diamond Price Prediction

A Machine Learning Analysis of Diamond Pricing Drivers and Predictive Modeling

Dataset	53,940 diamonds — carat, cut, color, clarity, dimensions, price
Objective	Identify price drivers and build a regression model to predict diamond price
Models Compared	Linear Regression, Ridge, Lasso, Elastic Net
Final Model	Linear Regression (highest R ² on test data)

Executive Summary

This project analyzes the classic 53,940-record diamonds dataset to understand which physical and quality attributes drive diamond price, and to build a regression model that predicts the price of a new diamond from its specifications. After cleaning (146 duplicate records removed, plus price outliers via the IQR method), four regression models were trained and compared. **Linear Regression** performed best, explaining 87.3% of price variance (R²) on unseen test data, with a typical prediction error (RMSE) of about \$967.

Records Analyzed	53,940	Best Model	Linear Regression
Records After Cleaning	50,400	Test R ² / RMSE	0.873 / \$967
Strongest Price Driver	Carat (r = 0.92)	New Diamond Prediction	\$6,803.60

1. Data Understanding & Exploratory Analysis

The dataset has 10 attributes: **carat** (weight), **cut**, **color** (D best–J worst), **clarity** (IF best–I1 worst), **depth** and **table** (proportion %), physical dimensions **x**, **y**, **z** (mm), and **price** (USD, target). There were no missing values, but 146 duplicate records were removed, and price outliers were removed using the IQR rule, leaving 50,400 records for modeling.

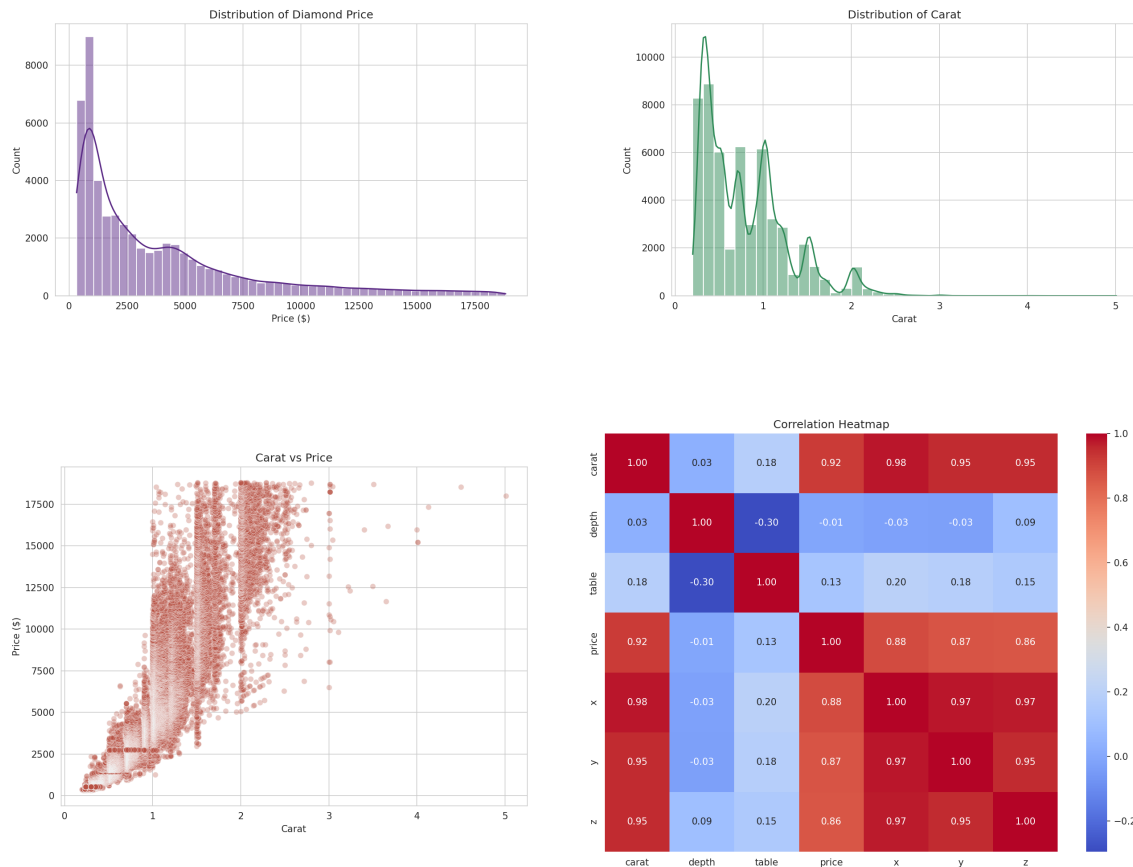


Figure 1. (top-left) Price distribution — right-skewed, most diamonds under \$5,000. (top-right) Carat distribution — concentrated between 0.2–1.5 carats. (bottom-left) Carat vs. price — strong positive, non-linear relationship. (bottom-right) Correlation heatmap of numeric features.

Q1. Which feature has the greatest impact on diamond price?

Carat has by far the strongest correlation with price ($r = 0.92$), making it the single most influential feature. Physical dimensions **x**, **y**, **z** are also strongly correlated ($r \approx 0.86$ – 0.88) since bigger stones are physically larger. Table and depth show weak correlation with price — cut proportions alone don't drive price the way size does.

Q2. Does a higher carat always result in a higher price?

Generally yes — carat and price move strongly together — but not always at the individual-diamond level. At the same carat weight, price still varies considerably by cut, color, and clarity, so carat is the dominant driver but not the only one.

2. Price by Quality Grade (Cut, Color, Clarity)

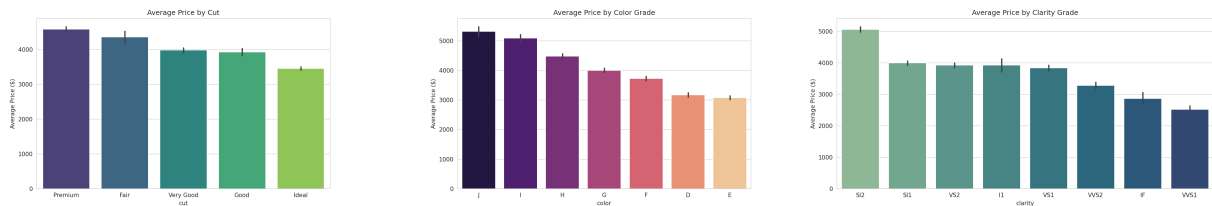


Figure 2. Average price by cut, color, and clarity grade.

Q3. Highest-price cut	Premium (\$4,584 avg)
Q4. Color linked to higher price	J — most tinted grade (\$5,324 avg)
Q5. Clarity contributing most to price	SI2 — lower clarity grade (\$5,063 avg)

These results are counter-intuitive at first glance: Premium cut, J color, and SI2 clarity are not the "best" grades, yet they show the highest average prices. This is **carat-weight confounding** — in this dataset, larger and heavier stones happen to more often carry these particular grades, which pulls their group averages up. Holding carat constant, the diamond market's usual premium for better cut/color/clarity still applies. The correlation analysis (Q1) and the regression model (Section 3), which account for all features together, give a more reliable read on each feature's true, independent effect on price.

3. Model Building & Evaluation

Categorical features (cut, color, clarity) were label-encoded and all features standardized. The cleaned data (50,400 records) was split 80/20 into train/test sets. Four regression models were trained and evaluated on the held-out test set.

Q6. Which regression model gives the best prediction performance?

Model	MAE	RMSE	R ² Score	Adj. R ²
Linear Regression	\$651	\$967	0.8732	0.8731
Ridge Regression	\$651	\$967	0.8732	0.8731
Lasso Regression	\$651	\$967	0.8732	0.8731
Elastic Net	\$721	\$1,006	0.8628	0.8626

Linear Regression achieved the best R² and lowest error, though it was nearly identical to Ridge and Lasso — regularization added little value with only 9 predictors and limited multicollinearity. Elastic Net underperformed, likely because its default L1/L2 mix over-shrinks coefficients for this data.

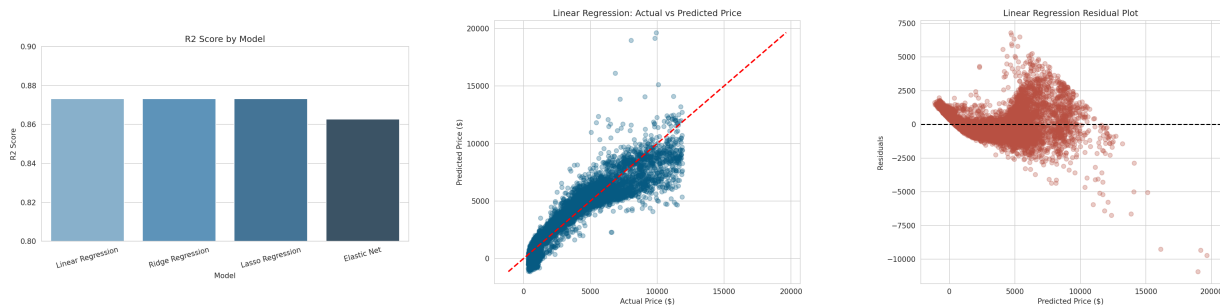


Figure 3. (left) R² by model. (center) Actual vs. predicted price, Linear Regression. (right) Residual plot — errors widen for higher-priced diamonds, meaning the model is less precise at the top end of the market.

4. New Diamond Prediction

Q7. Predicted price for a new diamond with the following specifications:

Carat	1.20	Depth	61.8
Cut	Premium	Table	57
Color	G	x (mm)	6.82
Clarity	VS2	y / z (mm)	6.79 / 4.20

Predicted Price **\$6,803.60**

Using the trained Linear Regression model, this 1.20-carat Premium-cut, G-color, VS2-clarity diamond is predicted at approximately **\$6,803.60**. Given the model's test RMSE of \$967, the true market price would typically fall within roughly $\pm \$967$ of this estimate.

5. Conclusion & Recommendations

- Carat weight is overwhelmingly the strongest single driver of diamond price; any pricing tool should weight it most heavily.
- Cut, color, and clarity have real but secondary effects — their raw group averages are confounded by carat and should be read alongside it, not in isolation.
- Linear Regression offers the best balance of accuracy and interpretability here; tree-based ensembles could be explored to capture non-linear interactions and potentially beat the 87.3% R^2 achieved.
- Error grows for very high-priced diamonds (see residual plot), suggesting a separate model or added features (certification, fluorescence) may help for luxury-tier stones.